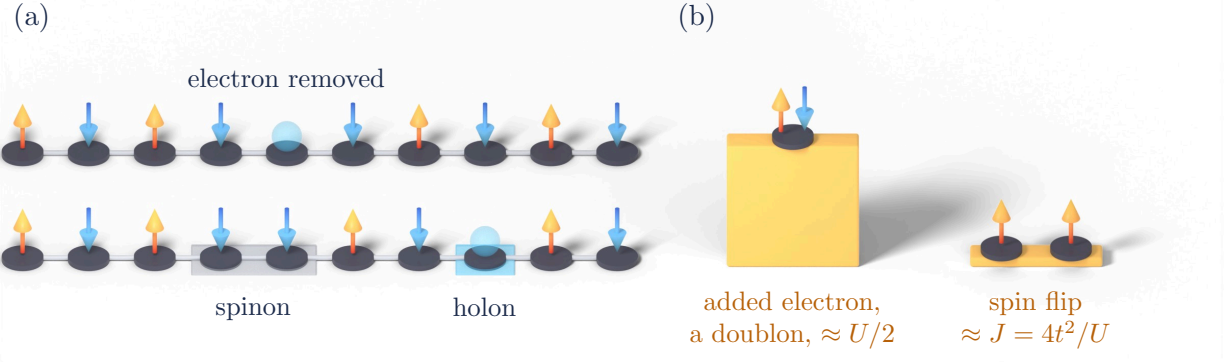

(a)
electron removed
spinon
holon
(b)
added electron,
a doublon, $\approx U/2$
spin flip
$\approx J = 4t^2/U$